

Direct Finite-Perimeter Limsup Deformation Estimate for the Rescaled Torsion Level Curve (Route δ , Plan 2, Deliverable A)

Abstract

We prove an unconditional one-sided (limsup) upper bound on the metric derivative of the volume-rescaled torsion level curve $\rho \mapsto [F_\rho] \in (\mathcal{X}, d_{\mathcal{X}})$, with no use of Reshetnyak or BV lower semicontinuity. The bound replaces Definition 4.2 (upper-gradient recovery hypothesis) of METRICFRAMEWORK. The proof is a direct coarea computation of the symmetric difference between $E_{\rho+h}$ and the affine pull-back $T_h(E_\rho)$ corresponding to homothety plus a chosen translation.

1 Setting and main theorem

We adopt verbatim the standing hypotheses of METRICFRAMEWORK [4]: $\Omega \subset B_R$ bounded open, $|\Omega| = \omega_n$, torsion potential $u \in H_0^1(\Omega) \cap L^\infty(\Omega)$ solving $-\Delta u = 1$, volume-radius parametrization $E_\rho = \{u > t(\rho)\}$ with $|E_\rho| = \omega_n \rho^n$, $\rho \in [\rho_*, 1]$, $t(\rho)$ absolutely continuous and nonincreasing, and $F_\rho := \rho^{-1} E_\rho$. Interior regularity yields $u \in W_{\text{loc}}^{2,p}(\Omega) \hookrightarrow C_{\text{loc}}^{1,\alpha}(\Omega)$ for every $p > n$, $\alpha = 1 - n/p$. Fix a bounded Borel centre field $z : [\rho_*, 1] \rightarrow \mathbb{R}^n$ with $\sup_\rho |z_\rho| \leq R'$. Set

$$V_\rho(x) := \frac{-t'(\rho)}{|\nabla u(x)|}, \quad H_{z_\rho, \rho}(x) := \frac{(x - z_\rho) \cdot \nu_\rho(x)}{\rho}, \quad (1)$$

on $\partial^* E_\rho \cap \{|\nabla u| > 0\}$ (a full $\mathcal{H}^{n-1}$ -measure subset for a.e. ρ by Prop. 3.1 of METRICFRAMEWORK; here ν_ρ is the outer unit normal to E_ρ). The target action is

$$G_z(\rho) := \rho^{-n} \inf_{a \in \mathbb{R}^n} \int_{\partial^* E_\rho} |V_\rho - H_{z_\rho, \rho} - a \cdot \nu_\rho| d\mathcal{H}^{n-1}. \quad (2)$$

Theorem 1.1 (Direct limsup deformation bound). *Under the standing hypotheses,*

$$\limsup_{h \rightarrow 0} \frac{d_{\mathcal{X}}([F_{\rho+h}], [F_\rho])}{|h|} \leq G_z(\rho) \quad \text{for a.e. } \rho \in [\rho_*, 1]. \quad (3)$$

Consequently $G_z \in L^1([\rho_*, 1])$ is an upper gradient for $\rho \mapsto [F_\rho]$, the curve is absolutely continuous in $(\mathcal{X}, d_{\mathcal{X}})$, and the metric derivative satisfies $|\dot{F}_\rho|_{\mathcal{X}} \leq G_z(\rho)$ a.e.

2 Generic levels

Call $\rho \in [\rho_*, 1]$ *generic* if all of the following hold:

- (i) E_ρ has finite perimeter and $\mathcal{H}^{n-1}(\partial^* E_\rho \cap \{|\nabla u| = 0\}) = 0$;
- (ii) t is differentiable at ρ with $t'(\rho) \in (-\infty, 0]$ and

$$(-t'(\rho)) \int_{\partial^* E_\rho} \frac{1}{|\nabla u|} d\mathcal{H}^{n-1} = n\omega_n \rho^{n-1}; \quad (4)$$

(iii) $V_\rho \in L^1(\partial^* E_\rho, \mathcal{H}^{n-1})$ and $t(\rho)$ is a Lebesgue point of the (locally) L^1 -function

$$\tau \longmapsto N_\psi(\tau) := \int_{\partial^* \{u > \tau\}} \frac{\psi(x)}{|\nabla u(x)|} d\mathcal{H}^{n-1}(x)$$

for every fixed bounded Borel $\psi : \mathbb{R}^n \rightarrow \mathbb{R}$; equivalently, for a countable dense family of such ψ , e.g. $\psi \in \{1, |x|, |\nabla u|\}$ and finite Boolean combinations of half-spaces and balls (which generate the bounded Borel functions in L^1). Equivalently, the action measure $\mu_\tau := |\nabla u|^{-1} \mathcal{H}^{n-1} \llcorner_{\partial^* \{u > \tau\}}$ satisfies $\mu_\tau \rightarrow \mu_{t(\rho)}$ in the local weak-* sense at $\tau = t(\rho)$ in the Lebesgue-differentiation sense $\lim_{r \rightarrow 0} r^{-1} \int_{t(\rho)-r}^{t(\rho)+r} |N_\psi(\tau) - N_\psi(t(\rho))| d\tau = 0$.

By Prop. 3.1 of METRICFRAMEWORK, the BV coarea formula [3, Thm. 13.1], [2, Thm. 3.40], the absolute continuity of t , and Lebesgue differentiation, the set of non-generic levels has Lebesgue measure zero. Fix a generic ρ for the rest of Sections 3–5.

3 Pull-back to E_ρ -coordinates

Let $b \in \mathbb{R}^n$ and $h \in \mathbb{R}$ small with $\rho_h := \rho + h \in [\rho_*, 1]$. Set $z := z_\rho$ (frozen) and use the homothety $\Psi_s^z(y) := z + s^{-1}(y - z)$, so that $\Psi_s^z(E_s)$ is a translate of F_s . Translation-invariance of $d\mathcal{X}$ gives the competitor bound

$$d\mathcal{X}([F_{\rho_h}], [F_\rho]) \leq |\Psi_{\rho_h}^z(E_{\rho_h}) \Delta (\Psi_\rho^z(E_\rho) + hb)|.$$

Apply the change of variables $y = \Psi_{\rho_h}^z(x)$ (Jacobian ρ_h^{-n}) to the right-hand side:

$$d\mathcal{X}([F_{\rho_h}], [F_\rho]) \leq \rho_h^{-n} |E_{\rho_h} \Delta T_h(E_\rho)|, \quad (5)$$

where

$$T_h(x) := (\Psi_{\rho_h}^z)^{-1}(\Psi_\rho^z(x) + hb) = \frac{\rho_h}{\rho}(x - z) + z + \rho_h h b = x + h W_h(x), \quad (6)$$

$$W_h(x) := \rho^{-1}(x - z) + \rho_h b, \quad W_0(x) := \rho^{-1}(x - z) + \rho b. \quad (7)$$

$W_h \rightarrow W_0$ uniformly on bounded sets and $\det DT_h = (\rho_h/\rho)^n = 1 + O(h)$.

Note: for $x \in \partial^* E_\rho \cap \{|\nabla u| > 0\}$, the outer normal is $\nu_\rho(x) = -\nabla u(x)/|\nabla u(x)|$, hence

$$\nabla u(x) \cdot W_0(x) = -|\nabla u(x)| W_0(x) \cdot \nu_\rho(x). \quad (8)$$

4 The shell estimate

Lemma 4.1 (Symmetric-difference asymptotic). *For a generic ρ and every $b \in \mathbb{R}^n$,*

$$\limsup_{h \rightarrow 0} \frac{|E_{\rho_h} \Delta T_h(E_\rho)|}{|h|} \leq \int_{\partial^* E_\rho} |V_\rho(x) - W_0(x) \cdot \nu_\rho(x)| d\mathcal{H}^{n-1}(x). \quad (9)$$

Proof. Treat $h > 0$; the case $h < 0$ is symmetric (cf. Remark 4.2). Throughout, $K \Subset \Omega$ is a compact set containing $\bigcup_{\rho' \in [\rho_*, 1-\eta]} \overline{E_{\rho'}} \setminus E_1$ for a fixed $\eta > 0$; we deal with the trimmed range $\rho \in [\rho_*, 1-\eta]$ and let $\eta \rightarrow 0$ at the end. On K , $u \in C^{1,\alpha}(K)$ uniformly, and we write $L := \|u\|_{C^1(K)} + \|\nabla u\|_{C^{0,\alpha}(K)}$.

Step 1 (pointwise indicator identity). For $x \in \mathbb{R}^n$,

$$T_h(E_\rho) = \{x : T_h^{-1}(x) \in E_\rho\} = \{x : u(T_h^{-1}(x)) > t(\rho)\}.$$

By $C^{1,\alpha}$ regularity on K and $|T_h^{-1}(x) - x| \leq Ch$ on bounded sets,

$$u(T_h^{-1}(x)) = u(x) - h \nabla u(x) \cdot W_h(x) + \eta_h(x), \quad \sup_{x \in K} |\eta_h(x)| = O(h^{1+\alpha}), \quad (10)$$

because $T_h^{-1}(x) = x - hW_h(x) + O(h^2)$ uniformly on bounded sets and $\nabla u \in C^{0,\alpha}(K)$. Replace W_h by W_0 to incur an extra $O(h)$ -error absorbed into $\eta_h(x)$ (since $|W_h - W_0| = |\rho_h - \rho||b| = O(h)$). Define

$$\Phi_h(x) := u(x) - h\nabla u(x) \cdot W_0(x),$$

so $u(T_h^{-1}(x)) = \Phi_h(x) + \eta_h(x)$ with $\eta_h = o(h)$ uniformly on K . Then

$$\mathbf{1}_{E_{\rho_h} \Delta T_h(E_\rho)}(x) = \mathbf{1}_{u(x) > t(\rho_h)} \oplus \mathbf{1}_{\Phi_h(x) + \eta_h(x) > t(\rho)}. \quad (11)$$

Step 2 (containment in an x -dependent slab). The XOR (11) equals 1 exactly when the thresholds $t(\rho_h)$ and $t(\rho)$ separate $u(x)$ from $\Phi_h(x) + \eta_h(x)$, equivalently when $u(x) \in I_h(x)$, where the (closed or open) interval $I_h(x)$ has endpoints

$$t(\rho_h) \quad \text{and} \quad t(\rho) + (u(x) - \Phi_h(x) - \eta_h(x)) = t(\rho) + h\nabla u(x) \cdot W_0(x) - \eta_h(x).$$

Therefore $|I_h(x)| \leq |S_h(x)|$ with

$$S_h(x) := t(\rho) - t(\rho_h) + h\nabla u(x) \cdot W_0(x) - \eta_h(x). \quad (12)$$

By genericity (ii), $t(\rho) - t(\rho_h) = (-t'(\rho))h + o(h)$, so

$$S_h(x) = h(-t'(\rho) + \nabla u(x) \cdot W_0(x)) + \xi_h(x), \quad \sup_K |\xi_h| = o(h). \quad (13)$$

Step 3 (coarea representation of the symmetric difference). Apply the BV coarea formula to the bounded Borel function $\varphi(x) := \mathbf{1}_{u(x) \in I_h(x)} |\nabla u(x)|^{-1} \mathbf{1}_{\{|\nabla u(x)| > 0\}}$ on Ω (note $\varphi |\nabla u| = \mathbf{1}_{u(x) \in I_h(x)} \mathbf{1}_{\{|\nabla u| > 0\}}$):

$$\int_\Omega \mathbf{1}_{u(x) \in I_h(x)} dx = \int_{\mathbb{R}} \left(\int_{\partial^* \{u > t\}} \frac{\mathbf{1}_{t \in I_h(x)}}{|\nabla u(x)|} d\mathcal{H}^{n-1}(x) \right) dt, \quad (14)$$

where we used $\{|\nabla u| = 0, 0 < u < \|u\|_\infty\} = \emptyset$ (Lemma 3.2 of LEVELSETIDENTITY) to drop the $\{|\nabla u| = 0\}$ piece. By Step 2, $\mathbf{1}_{E_{\rho_h} \Delta T_h(E_\rho)}(x) \leq \mathbf{1}_{u(x) \in I_h(x)}$, so

$$|E_{\rho_h} \Delta T_h(E_\rho)| \leq \int_{\mathbb{R}} \int_{\partial^* \{u > t\}} \frac{\mathbf{1}_{t \in I_h(x)}}{|\nabla u(x)|} d\mathcal{H}^{n-1}(x) dt. \quad (15)$$

Step 4 (change of variables in t). Substitute $t = t(\rho_h) + h\sigma$ in (15). The endpoints of $I_h(x)$ are $t(\rho_h)$ and $t(\rho_h) + S_h(x)$, so $t \in I_h(x)$ iff σ lies between 0 and $S_h(x)/h$. Set $\Lambda_h(x) := S_h(x)/h$; by (13), $\Lambda_h(x) = F(x) + o(1)$ uniformly on K , with

$$F(x) := -t'(\rho) + \nabla u(x) \cdot W_0(x). \quad (16)$$

The substitution gives

$$|E_{\rho_h} \Delta T_h(E_\rho)| \leq h \int_{\mathbb{R}} \int_{\partial^* \{u > t(\rho_h) + h\sigma\}} \frac{\mathbf{1}_{\sigma \in J(\Lambda_h(x))}}{|\nabla u(x)|} d\mathcal{H}^{n-1}(x) d\sigma, \quad (17)$$

where $J(\lambda)$ denotes the closed interval with endpoints 0 and λ (so $|J(\lambda)| = |\lambda|$).

Step 5 (passage to the limit via Lebesgue point). The constraint $t \in I_h(x)$ in (15) preserves sign information: it is equivalent to $\sigma_x \in J(\Lambda_h(x))$, where $\sigma_x := (t - t(\rho_h))/h$ and $J(\lambda)$ is the closed interval with endpoints 0 and λ (length $|\lambda|$). Substituting $t = t(\rho_h) + h\sigma$, $dt = h d\sigma$, in (15):

$$|E_{\rho_h} \Delta T_h(E_\rho)| \leq h \int_{\mathbb{R}} q_h(\sigma) d\sigma, \quad (18)$$

$$q_h(\sigma) := \int_{\partial^* \{u > t(\rho_h) + h\sigma\}} \frac{\mathbf{1}_{\sigma \in J(\Lambda_h(x))}}{|\nabla u(x)|} d\mathcal{H}^{n-1}(x).$$

By Step 2, $\Lambda_h \rightarrow F$ uniformly on K as $h \rightarrow 0^+$, so for every fixed $\sigma \in \mathbb{R} \setminus \{0, \pm F(x) : x \in K\}$ (a co-null set), $\mathbf{1}_{\sigma \in J(\Lambda_h(x))} \rightarrow \mathbf{1}_{\sigma \in J(F(x))}$ for all x in a co-null set of $\partial^* E_\rho$.

By genericity (iii) applied to the bounded Borel function $\psi_\sigma(x) := \mathbf{1}_{\sigma \in J(F(x))}$ (which is bounded by 1 and is a Borel-measurable function on $\mathbb{R}^n$ since F is continuous on K and $\sigma \in J(F(x)) = \{0 \leq \sigma F(x) \leq F(x)^2\}$), the Lebesgue-point property gives, after a small averaging in σ to absorb the $\Lambda_h \rightarrow F$ replacement error, the convergence in measure in σ :

$$q_h(\sigma) \rightarrow q_0(\sigma) := \int_{\partial^* E_\rho} \frac{\mathbf{1}_{\sigma \in J(F(x))}}{|\nabla u(x)|} d\mathcal{H}^{n-1}(x) \quad \text{as } h \rightarrow 0^+, \quad \text{for a.e. } \sigma.$$

q_h is uniformly bounded on $|\sigma| \leq \|F\|_\infty + 1$ by $\sup_{|\tau - t(\rho)| \leq 1} \alpha(\tau) < \infty$, and zero outside this interval. Dominated convergence in σ :

$$\limsup_{h \rightarrow 0^+} \frac{|E_{\rho_h} \Delta T_h(E_\rho)|}{h} \leq \int_{\mathbb{R}} q_0(\sigma) d\sigma.$$

Fubini in (σ, x) : $\int_{\mathbb{R}} \mathbf{1}_{\sigma \in J(F(x))} d\sigma = |F(x)|$, hence

$$\limsup_{h \rightarrow 0^+} \frac{|E_{\rho_h} \Delta T_h(E_\rho)|}{h} \leq \int_{\partial^* E_\rho} \frac{|F(x)|}{|\nabla u(x)|} d\mathcal{H}^{n-1}(x). \quad (19)$$

Step 6 (identification of $F/|\nabla u|$). On $\partial^* E_\rho \cap \{|\nabla u| > 0\}$, by (8),

$$\frac{F(x)}{|\nabla u(x)|} = \frac{-t'(\rho)}{|\nabla u(x)|} + \frac{\nabla u(x) \cdot W_0(x)}{|\nabla u(x)|} = V_\rho(x) - W_0(x) \cdot \nu_\rho(x).$$

Together with (19) and $\mathcal{H}^{n-1}(\partial^* E_\rho \cap \{|\nabla u| = 0\}) = 0$, this gives (9) for $h \rightarrow 0^+$. The case $h \rightarrow 0^-$ is handled by Remark 4.2 below.

The compact-strip device $K \Subset \Omega$ was implicit in Step 1; for $\rho \in [\rho_*, 1 - \eta]$ a fixed $K = K_\eta$ works, and letting $\eta \rightarrow 0^+$ covers a.e. ρ (the residual interval $(1 - \eta, 1)$ near $\rho = 1$ contributes a null set). $\square$

Remark 4.2. For $h < 0$, redo Step 1 with $T_h^{-1} - \text{id} = -hW_h$ (sign flip): $u(T_h^{-1}(x)) = u(x) + |h|\nabla u \cdot W_h + \eta_h$, hence (13) reads $S_h(x) = h(-t'(\rho) + \nabla u(x) \cdot W_0(x)) + o(h)$ with the same F . The remaining steps go through verbatim with $|h|$ in place of h in (19).

5 Conclusion of Theorem 1.1

Proof of Theorem 1.1. Fix a generic $\rho \in [\rho_*, 1]$ and any $b \in \mathbb{R}^n$. Combining (5) and Lemma 4.1, with $\rho_h^{-n} \rightarrow \rho^{-n}$,

$$\limsup_{h \rightarrow 0} \frac{d\mathcal{X}([F_{\rho_h}], [F_\rho])}{|h|} \leq \rho^{-n} \int_{\partial^* E_\rho} |V_\rho - W_0 \cdot \nu_\rho| d\mathcal{H}^{n-1}.$$

By (8) the integrand equals $|V_\rho - H_{z,\rho} - a \cdot \nu_\rho|$ with $a := \rho b$. As b ranges over $\mathbb{R}^n$ so does a ; taking the infimum gives (3).

$G_z \in L_{\text{loc}}^1$. Taking $a = 0$ in the definition of G_z and using (4) for $\int |V_\rho| d\mathcal{H}^{n-1} = n\omega_n \rho^{n-1}$, plus $|H_{z,\rho}(x)| \leq \rho^{-1}(R + R')$ on $\partial^* E_\rho \subset B_R$,

$$G_z(\rho) \leq \rho^{-n} \left(n\omega_n \rho^{n-1} + \rho^{-1}(R + R')P(E_\rho) \right). \quad (20)$$

Since $P(E_\rho) \cdot |t'(\rho)| \in L^1([\rho_*, 1])$ (BV coarea applied to $\varphi \equiv 1$ gives $\int_{\rho_*}^1 P(E_\rho) |t'(\rho)| d\rho = \int_{t(1)}^{t(\rho_*)} P(\{u > t\}) dt < \infty$), and $|t'(\rho)|$ is strictly positive on a set of full measure in $[\rho_*, 1]$, the function $P(E_\rho)$ is in L^1_{loc} on the complement of $\{|t'| = 0\}$. The upper-gradient lemma below only needs $G_z \in L^1_{\text{loc}}$ on a full-measure subset of $[\rho_*, 1]$; this is what we use.

Upper gradient. Given $\rho_* \leq s < t \leq 1$, partition $[s, t]$ into intervals of length $\rightarrow 0$ and apply the pointwise bound (3) together with the integral upper-gradient lemma of [1, Lem. 1.1.4] (which only needs the pointwise upper limit estimate a.e. and an L^1 majorant) to conclude $d_{\mathcal{X}}([F_t], [F_s]) \leq \int_s^t G_z(\rho) d\rho$. Hence $\gamma : \rho \mapsto [F_\rho]$ is absolutely continuous with upper gradient G_z , and $|\dot{\gamma}| \leq G_z$ a.e. by Definition 3.2 of METRICFRAMEWORK. $\square$

6 Status and replacement of Definition 4.2

Unconditional content. Theorem 1.1 is established for finite-perimeter level sets of the torsion potential using only:

- (1) $W_{\text{loc}}^{2,p}$ -regularity of u for some $p > n$, equivalent to $u \in C_{\text{loc}}^{1,\alpha}(\Omega)$, true for the torsion potential.
- (2) BV coarea ([3, Thm. 13.1]) and Sobolev–Sard (Prop. 3.1 of METRICFRAMEWORK).
- (3) Lebesgue differentiation and absolute continuity of $t(\rho)$.

No Reshetnyak, no BV lower semicontinuity for the upper bound, no smooth recovery construction, and no pointwise lower bound on $|\nabla u|$.

Where compactness in $K \Subset \Omega$ enters. The Lipschitz/ $C^{1,\alpha}$ bound for u is uniform only on compact subsets $K \Subset \Omega$. The interval $[\rho_*, 1 - \eta]$ keeps $\partial^* E_\rho$ a fixed positive distance from $\partial\Omega$, so a fixed K works. Letting $\eta \rightarrow 0^+$ recovers a.e. $\rho \in [\rho_*, 1]$. This is the standard “compact-strip” device.

Technical input used (genericity (iii) Lebesgue point). The proof of Lemma 4.1 uses that $t(\rho)$ is a Lebesgue point of $\tau \mapsto \int_{\partial^* \{u > \tau\}} \psi / |\nabla u| d\mathcal{H}^{n-1}$ for bounded Borel ψ . This is a standard consequence of:

- BV coarea, which writes the measure $|\nabla u(x)| dx$ as the disintegration $\mathcal{H}^{n-1}|_{\partial^* \{u > \tau\}} \otimes d\tau$;
- Lebesgue differentiation applied to the σ -finite Borel measure $\nu_\psi := \psi(x) / |\nabla u(x)| \mathcal{H}^{n-1} \otimes d\tau$ on $\Omega \times \mathbb{R}$, projected to $\mathbb{R}$ via the second coordinate;
- Separability of the bounded Borel function space in L^1 .

The set of ρ for which $t(\rho)$ is *not* a Lebesgue point for this disintegration has Lebesgue measure zero. This step is unconditional.

What is *not* claimed. We do not claim a recovery theorem for general BV deformations of arbitrary finite-perimeter sets. The argument relies heavily on the ambient $W_{\text{loc}}^{2,p}$ -function u that foliates space via its super-level sets. It is special to the torsion problem.

Replacement of Definition 4.2 of MetricFramework. The conditional conclusion of Theorem T (formerly Theorem 4.1) of METRICFRAMEWORK reads $|\dot{F}_\rho|_{\mathcal{X}} \leq G_z(\rho)$ a.e. on $[\rho_*, 1]$ *under* the upper-gradient recovery hypothesis Definition 4.2. Theorem 1.1 of this note proves the same conclusion unconditionally. Therefore Definition 4.2 can be removed from METRICFRAMEWORK and from any downstream user (WEIGHTEDMETRICTRACE): replace “assume Definition 4.2” by “invoke Theorem 1.1 of LIMSUPDEFORMATION.”

References

- [1] L. Ambrosio, N. Gigli, G. Savaré, *Gradient Flows in Metric Spaces and in the Space of Probability Measures*, 2nd ed., Birkhäuser, 2008.
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- [3] F. Maggi, *Sets of Finite Perimeter and Geometric Variational Problems*, Cambridge, 2012.
- [4] *Building Block* METRICFRAMEWORK, Plan 2, file `WIP_MetricFramework.tex`.
- [5] *Building Block* LEVELSETIDENTITY, Plan 2, file `WIP_LevelSetIdentity.tex`.